



How to: Build your own EntoFARM



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<https://www.ento.farm/>

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GLOSSARY



Frass: The residues of insect-rearing that mainly consist of insect faeces, exuviae and feed residues and can be used as soil fertilizer.



Larva: The juvenile stadium of holometabolous insects. There are several larval stages, so-called instars, in the life cycle of mealworms before metamorphosis (transforming them to adult beetles).



Pupa: During pupation, the metamorphosis from larva to adult beetle happens.

SUMMARY

This **guide** provides a simple, educational approach to raising **mealworms** while teaching **sustainability** and **food system resilience**. It is ideal for schools, families, and urban farms!

This guide aims to teach **students** and **educators** how to build and manage a mealworm farm to promote **circular economy principles, sustainability, and hands-on learning**. It supports environmental education through practical insect farming activities.

Mealworms (larvae of *Tenebrio molitor*) help close the **loop** in the circular economy by consuming **food waste** (organic scraps), producing **protein-rich biomass** (for animal or human consumption), creating **frass** (a nutrient-rich fertilizer) and requiring fewer resources than traditional livestock.

Within this manual we describe what is needed to start a mealworm farm, in terms of **equipment** and **tools** [trays/crates, feeding substrate (dry and wet) sieves etc.], whereas we guide the students and their teachers through the process of running their own mealworm farm [mealworm larvae feeding and harvesting, adult breeding, frass collection etc.].

Hope you enjoy!

WHAT IS ENTOFARM ABOUT?

The primary objective of the **EntoFARM project** is to enhance environmental awareness and understanding of circular economy practices among diverse groups of **secondary school students** by introducing innovative insect farming as a practical application of **sustainability**. More particularly, EntoFARM aims to promote interest in circular economy and sustainability strategies by engaging students in the design, development, and management of small-scale insect farms. Additionally, it aims to empower teachers by providing innovative resources, specifically a comprehensive educational toolkit on circular economy and insect farming, designed to strengthen their capacity to deliver engaging, up-to-date lessons on sustainability and effectively integrate these topics into their curricula.

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The Consortium



Fraud Line Enterprise Risk Management and Compliance Services (INNOFY/Fraud Line) (Coordinator, Athens, Greece, www.fraudline.gr)



University of Thessaly, Department of Agriculture, Crop Production and Rural Environment, (UTH) (Volos, Greece, www.agr.uth.gr)



University of Patras, Department of Agriculture (UPatras) (Messolonghi, Greece, <https://agriculture.upatras.gr/en>)



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Association for Internationalization of Education and Science (Stip, The Republic of North Macedonia, www.interedu.mk)

Contact Details and Social Media

You can find more information and updates regarding the project:



<https://www.ento.farm>



Ento FARM



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We are happy to share more resources, answer questions and support your journey in sustainable insect farming!

INTRODUCTION

Have you ever wondered what happens to all the food scraps we throw away?

What if we told you that tiny, amazing creatures could help us turn that waste into something useful? Get ready to discover the secret world of mealworms and learn how to build your very own mealworm farm!

The superpower of circular economy!

Imagine a world where nothing is wasted! That's the idea behind something called the **Circular Economy**.

Think about it like this:

- In a **linear economy**, we "take, make, and dispose." We grab resources, make stuff, use it, and then throw it away. This creates a lot of trash and uses up our planet's precious resources.
- In a **circular economy**, we "reduce, reuse, and recycle." Instead of throwing things away, we find ways to keep them in use for as long as possible. It's like a big, never-ending loop where resources are used again and again!

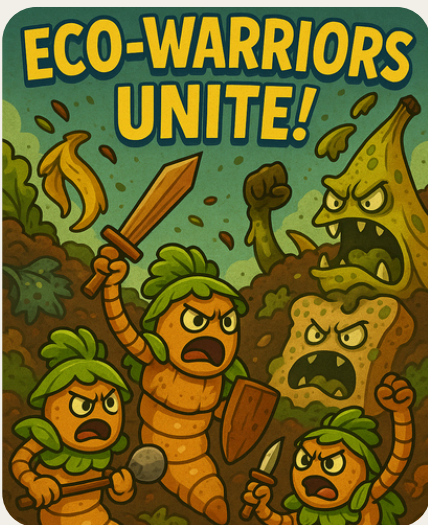


Why is this important?

Our planet has limited resources, and we're making too much trash. A circular economy helps us protect our environment, save energy, and create less pollution. It's all about being smart with what we have!



How insects can become circular economy heroes!



Now, where do our mealworm friends fit into this superpower plan?

Mealworms (which are actually the larvae of specific beetles called *Tenebrio molitor* or yellow mealworms) are incredible recyclers! They can eat many types of organic waste, like fruit and vegetable scraps, and turn them into valuable products.

Here is how they can help:

- **Waste Eaters:** Instead of food waste going to landfills where it creates harmful gases, mealworms munch on it, reducing the amount of trash.

- **Protein Powerhouses:** The mealworms themselves are packed with protein and healthy fats! They can be used as feed for animals (like chickens, fish, and even

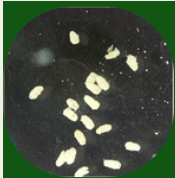


- some pets) or, in some cultures, for humans! This means we rely less on traditional livestock animals, which often require a lot of land and water.
- **Natural Fertilizer:** What the mealworms leave behind, called "frass" (it's basically their poop!), is a fantastic natural fertilizer for plants. So, they help our gardens grow too!
- **Less Pollution:** Raising mealworms uses much less water and land compared to farming animals like cows or pigs. They also produce fewer greenhouse gases. It's a win-win for the planet!

The amazing life of beetles: A four-stage journey

Have you ever seen a ladybug, a June bug, or even a mealworm?

These are all types of beetles, and they have an amazing secret: they go through a complete transformation in their lives! This special kind of development is called "complete metamorphosis," and it's why beetles, including our mealworm friends, have four major life stages: **egg, larva, pupa, and adult.**



Egg Stage: This is where it all begins! Female beetles lay tiny eggs, often in a safe spot where the future babies will have food when they hatch.



Larva Stage: Once the egg hatches, out comes the larvae! For mealworms, this is the "mealworm" stage you're familiar with. Larvae are constantly eating to grow big and store up energy. They shed their skin multiple times as they get larger, which is called molting.



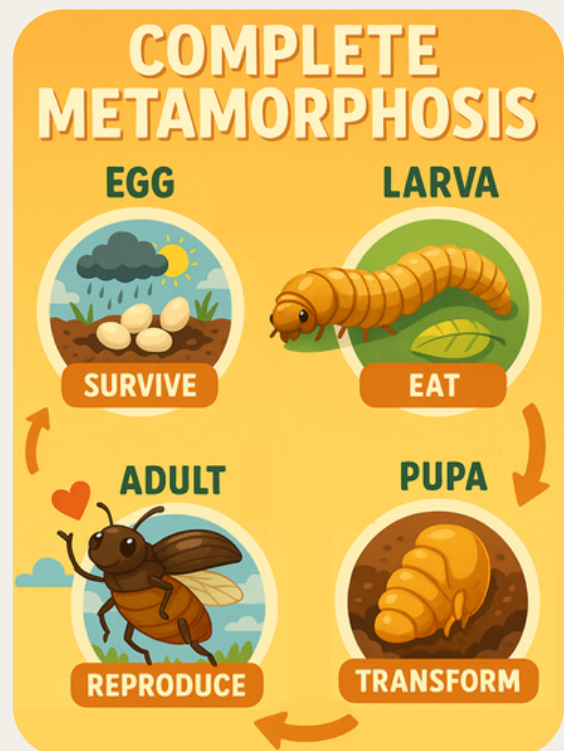
Pupa Stage: After the larva has eaten enough and grown to its full size, it enters the pupa stage. This is a resting stage where incredible changes happen inside, like a secret transformation. The pupa usually doesn't move or eat. Its purpose is to transform the insect from a worm-like larva into a winged adult.



Adult Stage: Finally, the pupa transforms into the adult beetle!
For mealworms, this is the darkling beetle.

Why four stages with different characteristics?

This complete metamorphosis is an incredibly clever strategy for beetles! Think about it: the eggs are quite resilient in extreme environmental conditions and so they can survive in any situation. The **larva** (like our mealworm) is a dedicated **eating and growing machine**. Its whole purpose is to consume as much food as possible to build up energy. The **adult beetle**, on the other hand, has a completely different job: **finding a mate and laying eggs** to create the next generation, and often dispersing to new locations. By having such distinct life stages, with different body forms and often different food sources and habitats, the young (larvae) don't have to compete directly with the adults for resources. This specialization allows them to be **super efficient** at different parts of their life, making them one of the most **successful groups of insects** on Earth!



It's a truly efficient way of life!!

THINGS YOU'LL NEED TO START YOUR OWN MEALWORM FARM

Ready to become a mealworm farmer?

Here's what you'll need to get started. You can find many of these items around your house or at a local store!

Trays / Crates

These are the "apartments" for your mealworms!

You'll need several shallow trays. Plastic storage bins and containers or old shoeboxes (lined with plastic) sourced from supermarkets or retailer stores could be used for mealworm rearing (Figure 1).



Figure 1: Plastic containers that can be used for mealworm rearing.

When using plastic containers with a lid, it is important to create a hole in the center of the plastic lid and cover it with fine mesh to ensure proper air circulation within the container (Figure 2).



Figure 2. Plastic container with a hole in the center of the lid covered with fine mesh to ensure proper air circulation within the container.



Look for stackable trays that can be placed on top of each other to save space and allow vertical expansion (Figure 3). Care should be taken to leave a slight space between each stacked container to ensure adequate air circulation. This helps with ventilation and keeps your mealworms happy!



Figure 3. Plastic containers stacked on top of each other to save space.

Of course, there are also dedicated trays / crates for mealworm rearing (Figure 4) that also work great and are stackable (Figure 5)!



Figure 4. Dedicated plastic trays for insect rearing.



Figure 5. Commercial stacked trays in a pilot scale mealworm production facility.

Feedstock / Feeding Substrate

This is the material your mealworms will feed on - it serves as both their food and their living substrate!



Wheat bran is a popular choice, even for commercial mealworm production.



Other amylaceous substrates, such as **stale bread crumbs**, can work as well.

As mealworm farmer, you can experiment with a huge variety of organic substrates as mealworm feed! Various agricultural **side-streams** and **by-products** have been successfully explored by UTH as feedstocks for mealworm rearing, e.g., barley, pea or oat by-products (Figure 8). You can also experiment with other organic substrates, as mealworm feed. Just be sure to grind any whole kernels, as the insects can't eat them in their whole form!



- **Important:** Make sure your feeding substrate is dry and free from mold.
- You'll need enough to create a layer about **3-5 cm** deep in each tray.



Figure 8. Various agricultural side-streams and by-products that have been successfully explored by UTH as feedstocks for mealworm rearing (left: as a whole, right: ground).

- A:** Barley by-product (beard hairs, husks and other non-seed materials),
- B:** Pea by-product (small, broken seeds),
- C:** Pea by-product (bean pods and other non-seed materials with few small and broken seeds) and
- D:** Oat by-product (beard hairs, husks, small seeds).

Moisture Source

Mealworms need a little moisture, but not too much!



Work perfectly as moisture source. These also provide extra nutrients to mealworms!



Very wet or moldy fruits/vegetables, as these can cause mold in your farm.

Mesh

A mesh can be placed on top of the oviposition substrate to prevent adult beetles from cannibalizing their eggs.



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Starter mealworms

You need some mealworms to begin your farm!



You can buy these from **pet stores** (they're often sold as reptile food) or online.



Start with a good number, like **500-1000 large larvae**, to ensure a healthy colony.



You can always contact University of Thessaly and University of Patras for any questions or to acquire your starter mealworms.

Environmental conditions monitoring

Mealworms thrive in specific conditions: a temperature of **25–28 °C** and a relative humidity of around **60%**. Maintaining these levels helps them grow faster, stay healthier, and reproduce more effectively.

While professional insect farms rely on **growth chambers** with precise controls, you don't need specialized equipment to get started. With a bit of creativity, you can set up and monitor the environment using everyday items you may already have at **home**, or by purchasing affordable equipment from **reptile shops**.



Temperature Control

Mealworms are happiest in a warm environment, but they don't like it too hot or too cold. Here are some ways to manage temperature:

Warming Options

- **Heating pads** (the kind used for reptiles or seed germination) placed under or beside the container.
- Incandescent or **LED heat lamps** that gently warm the area without overheating.
- **Sunny windows** (but be careful—direct sunlight can cause sudden overheating).
- **Warm spot** in your room near a radiator or heater, but not directly on it.

Cooling Options

- If your space gets too hot, you can place the container in a **cooler corner** of the room, away from direct sun.
- Small **desk fan** to keep the air moving and prevent overheating.
- In really hot weather, you can even use a **frozen water bottle** wrapped in a towel placed near the container to gently lower the temperature.

Humidity Control

Mealworms need moderate humidity (around 60%). Too dry, and they dehydrate, too wet, and mold grows. Here are simple tricks:

Raising Humidity

- Place a slightly **damp paper towel** in a corner of the container (not too wet, to avoid mold).
- Add slices of carrot, apple, or potato, which also serve as food and water sources.
- Keep the container covered with **breathable fabric** (like mesh or cotton cloth) to slow down moisture loss.

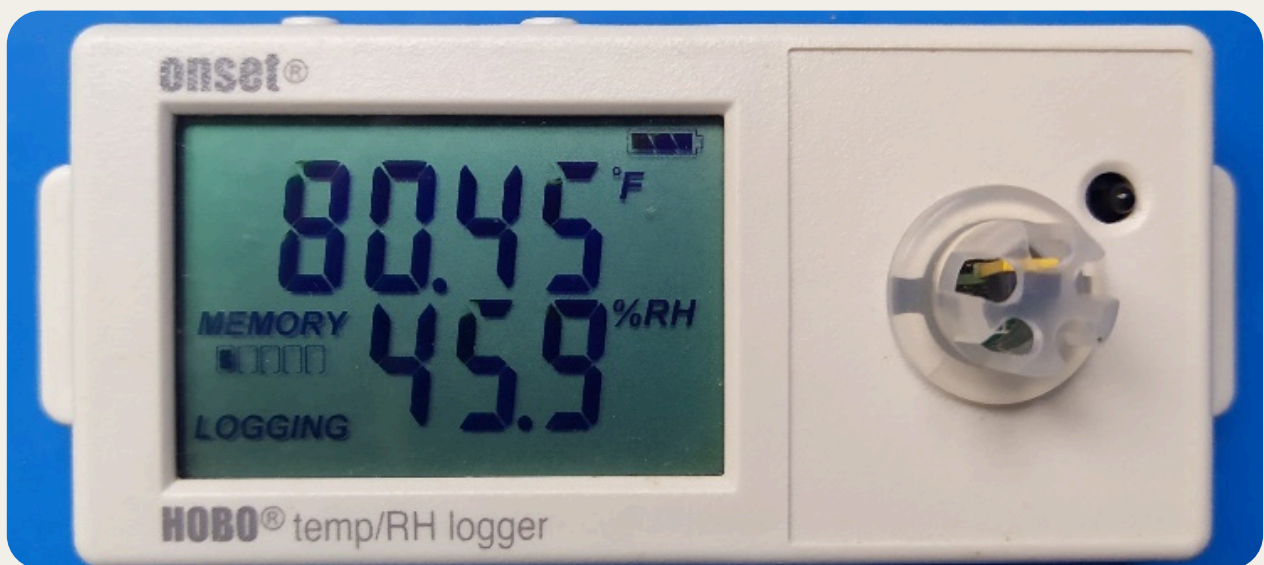
Lowering Humidity

- Remove extra **food scraps** if they start getting soggy.
- Add a bit of **dry oats** or bran to absorb excess moisture.
- **Improve airflow** by opening the cover more often or using a **small fan** nearby.

Monitoring Tools

Even though professional **data loggers** are ideal (see photo below), you can start with simple and affordable tools:

- **Digital thermometer-hygrometer** combos.
- **Aquarium thermometers** to keep an eye on temperature.
- Even a **Bluetooth sensor** with a a phone app.

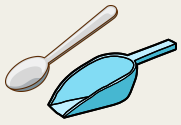


With these methods, you **don't need to invest** in expensive equipment right away. You can **experiment** with your mealworm farm, learn how the insects respond to changes, and slowly improve your setup as you go.

It's like being both a farmer and a scientist!

Tools

You will need:



Small and bigger **scoops and spoons** for handling feeding substrates and other materials needed for mealworm rearing.



Sieves (with a mesh size between 500 µm to 3-5 mm) for separating the different mealworm life stages, as well as mealworms from frass.



A **spray bottle** (for misting if humidity is too low, though often not needed with fresh produce).



Small or bigger **brushes** for cleaning.

Pest protection measures (optional but recommended)



Watch out for foreign insects (moths, beetles) on your farm (use insect traps for monitoring)! These are considered as pests and need to be removed from the trays!

HOW TO RUN YOUR MEALWORM FARM

Now that you have all your materials, let's get farming!

More information for the mealworm life cycle (see also pages 9-10):

1. **Egg:** Tiny, white eggs are laid by adult beetles. They hatch in about 4-12 days.
2. **Larva (Mealworm!):** This is the stage you will be most familiar with! They eat a lot and grow, shedding their skin (molting) several times. This stage can last from a few weeks to several months, depending mostly on the temperature and feeding substrate.
3. **Pupa:** After growing big enough, the larva transforms into a pupa. It looks like a hard, white, curled-up creature and doesn't move or eat. This stage lasts about 1-2 weeks.
4. **Adult (Darkling Beetle):** The pupa transforms into a beetle! At first, it's light brown, then turns dark brown or black. Adult beetles don't eat much, but they mate and lay eggs, starting the cycle all over again!

Setting up your farm

1. Prepare your trays:

- Clean all your trays thoroughly.
- For your main "larval" tray, add a 4-6 cm layer of your chosen feeding substrate (e.g., wheat bran).

2. Introduce your mealworms:

- Gently place your starter mealworm larvae onto the feeding substrate in the larva tray.
- You need to be careful of how much larvae you will place into each tray. A general rule is to place 2-4 larvae / cm².
- Add a few pieces of your moisture source (like carrot slices) on top of the bedding.

3. Find the right spot:

- Mealworms prefer a warm, dark, and slightly humid environment.
- Temperature: around 25-28 °C. Avoid direct sunlight or very cold spots.
- Humidity: Around 50-60% relative humidity is good. The moisture source usually provides enough humidity.
- Place your stacked trays in a quiet, undisturbed area.

Feeding your mealworms

(See also page 15)

Mealworms are not picky eaters, but a good diet helps them grow big and healthy!

Main Food (dry feed): Their rearing substrate (wheat bran, oats, etc.) is their primary source of food. They will eat it slowly as they burrow.

Moisture source (wet feed): Add fresh fruit and vegetable scraps (like carrot peels, apple cores, potato slices, lettuce leaves) 2-3 times a week.



Avoid: Citrus fruits, very watery items, or anything that has been treated with pesticides.



Important: Only add as much as they can eat in a day or two to prevent mold and pests. Remove any uneaten or moldy pieces promptly.

Farm Maintenance and Monitoring = Keeping Your Mealworms Happy!

Regular care is the secret to a healthy and productive mealworm farm! By following these simple steps for maintenance and harvesting, you'll have a thriving colony in short time.

Caring for your larval trays

- **Feeding Time!**

Your mealworms are hungry little eaters! You'll need to check their trays regularly to see how much food (like oats or bran, and fresh scraps like fruit and veggie pieces) they have left. Don't wait until it's completely empty! When you notice their food supply getting low, it's time to add more. Over time, you'll get a good feel for how much they gobble up each day – it's all part of becoming a mealworm expert!

- **Frass Removal – The Mealworm Treasure!**

As your mealworms munch away and grow, they leave behind tiny droppings called "frass". This looks a bit like fine sawdust and will pile up at the bottom of their tray. You can let the frass build up until you're ready to harvest your mealworms, or if the layer gets very thick, you can carefully use a sieve to separate the mealworms from their frass.



[Wheat bran (left) transformed into frass (right) by mealworms.]

Remember, this frass is pure gold for your plants – it's an excellent, nutrient-rich fertilizer!

Monitoring conditions



Temperature: Keep an eye on the temperature of your mealworm setup. If it's too cold, their growth will slow down significantly. If it's too hot, they might get stressed, try to escape, or even die. Aim for a comfortable room temperature (around 25-28°C).



Humidity: Regularly check their moisture source (like a slice of fruit or vegetable). If it dries out quickly, add more or mist lightly with water (but be careful not to make the feeding substrate soggy, as too much moisture can lead to mold!).



Smell: A healthy mealworm farm should smell earthy, a bit like oats or grain. A bad smell (sour, moldy, or rotten) is a warning sign that something is wrong, usually too much moisture or uneaten, decaying food. Address this quickly!

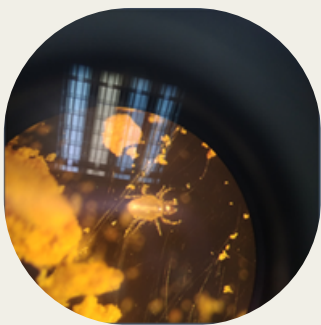


Pests: Occasionally check for any unwanted guests like mites or other tiny insects (see example pictures below). Good moisture management and regular cleaning are your best defenses against these pests!

Pests that can “visit” your mealworm farm.



Moths



Mites



Beetles



Psocids

Harvesting Your Mealworms: From Farm to Feast!

When you start noticing that **more than 10%** of the larvae have transformed into white, curled-up pupae appearing in your larvae tray (this usually happens **after 5-10 weeks**, depending on the conditions), it's a clear sign! Many of your mealworms are now mature and ready to be harvested, or you can collect the pupae to continue the exciting life cycle of your farm.



To harvest your larvae (mealworms):



- **Separate Them:** You can easily use a sieve to separate them from their feeding substrate and frass.
- **Fast Them:** If you plan to feed them to animals or use them yourself, it's recommended to

"fast" them for 24 hours by keeping them without food to clear their digestive system.

- **Harvest Them:** You can place them in a freezer for a few hours.
- **Store Them:** They can be stored in the refrigerator (they'll go dormant) for a few weeks or dried for longer storage in room temperatures (like any common flour).

Collecting Pupae and Transferring Adults (Maintaining Your Farm):

To keep your mealworm farm going strong, it's important to keep a steady supply of adult beetles (your "breeders"). Here's how you prepare the next generation:

- **Sieve Gently:** Once a week for about four weeks, use a sieve to carefully separate the pupae from the remaining larvae. This gradual process ensures you catch most of them.
- **Soft Landing:** As you sieve, make sure the pupae fall onto a soft surface, like a layer of fresh bran, so they don't get hurt. This substrate will also be their first meal when they become adult beetles!
- After about **four weeks**, any remaining larvae in that original tray can be removed, as most will have pupated.
- **Adult beetle Transfer:** After about two weeks, your pupae will have transformed into darkling beetles! Collect these new adult beetles.



- Transfer adult beetles to a new tray that has fresh oviposition substrate (like wheat bran) and a good moisture source (the moisture source should be evenly distributed across the tray and replaced when depleted.) This will be



your main "breeding" tray. It helps if you place a mesh on top of the oviposition substrate (Figure 23), making sure there is no gap between the mesh and the substrate. This will prevent adult beetles from cannibalizing their own eggs!

- The beetles will mate and lay tiny eggs in the substrate, starting the cycle all over again!
- Use these newly emerged adults to replace any old or dead adult beetles in your existing breeding trays, ensuring a continuously productive farm.

Managing your breeding tray for new generations:

This is where the magic of starting a new batch of mealworms happens!

Egg laying: Your adult beetles will be busy laying tiny, almost invisible eggs directly into the feeding substrate in their breeding tray.

Starting new larva trays: To get a new batch of mealworms, you don't collect individual eggs. Instead, after your adult beetles have been in a breeding tray for about 2-4 weeks, you'll transfer the adult beetles themselves to a brand-new breeding tray with fresh oviposition substrate.

- The substrate they leave behind in the previous breeding tray now contains all the tiny eggs!
- Gently set aside this "egg tray" in a quiet spot. After a few days to a couple of weeks (depending on conditions), these eggs will hatch into super tiny, almost microscopic larvae. You might not even see them at first, but they're there!
- This tray effectively becomes your new "larva starter" tray, and the cycle continues as these tiny larvae begin to grow!

Important tips for success!

Patience is the key: Mealworm farming takes a little time.

Observe your mealworms and learn from them.

Don't overfeed moisture: This is the most common mistake!

Too much moisture leads to mold and pests.



Keep it clean: Regular cleaning prevents problems and keeps your mealworms healthy.

Experiment: Try different food scraps (in small amounts) and see what your mealworms like best!

Have fun becoming a mealworm farming expert and contributing to a more circular world!



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